# A Semigroup Theory of Governance:

Operator Invariance, Bayesian Epistemics,
and Kernel Safety in Stochastic Systems

Usman Zafar, Ph.D.

Founder, Zulfr info@zulfr.com

March 2026

**Abstract**

The *Governance Framework* (GF), a unified measure-theoretic formulation of governance in open, adaptive, and partially observable systems is introduced. GF is built on a filtered probability space with Polish state and belief spaces, and synthesizes eight mathematical disciplines: measure-theoretic probability (Kolmogorov), functional analysis, $C_0$-semigroup theory, Bayesian inference, stochastic process theory, game theory, fixed-point theory, and constrained control theory.

System uncertainty is encoded as a probability measure on state space; evolution is governed by a stochastic operator semigroup; epistemic state is a Bayesian belief process on $\mathcal{P}(X)$; and multi-agent interaction is formalized through game-theoretic coupling in operator selection. Safety is defined as invariance of a *maximal forward-invariant kernel* $K^\star \subset Z$ under semigroup composition, unifying viability theory and robust control.

We prove five theorems with complete proofs: *(i)* safety kernel invariance under operator semigroup composition; *(ii)* existence and uniqueness of a consistent stochastic evolution via Kolmogorov extension; *(iii)* existence of a governance equilibrium as a fixed-point operator via Schauder's theorem; *(iv)* a Belief Contraction Lemma establishing $d_{\mathrm{TV}}$-convergence of Bayesian updates; and *(v)* an Operator Separation Theorem identifying the minimal operator sub-class that preserves safety under adversarial composition. A Metatheorem establishes that each of the eight constitutive disciplines is load-bearing: removing any one collapses at least one theorem.

GF unifies stochastic control, POMDPs, constrained RL, stochastic games, and organizational governance as formal special cases. The seven appendices supply unified notation, full proofs, a taxonomy of twenty-one cases, three TikZ system diagrams, a PSPACE verification result for finite models, a comparison against six frameworks, and three worked examples, including a safe-learning robotic manipulation instance.

**Keywords:** semigroup theory, stochastic operator systems, Bayesian epistemics, safety kernel invariance, viability theory, constrained reinforcement learning, partial observability, governance framework, PSPACE complexity, operator fixed points.

# Contents


**1 Introduction** **4**
- 1.1 Motivation . . . 4
- 1.2 The Operator-Level Shift . . . 4
- 1.3 Contributions . . . 4
- 1.4 Scope and Non-Goals . . . 5
- 1.5 Paper Organization . . . 5

**2 Mathematical Foundations** **5**
- 2.1 Topological and Measurable Spaces . . . 5
- 2.2 Filtrations and Information Structure . . . 6
- 2.3 Operator Structure . . . 6
- 2.4 Bayesian Belief Dynamics . . . 6
- 2.5 Game-Theoretic Coupling . . . 7
- 2.6 The Governance Framework — Unified Definition . . . 7

**3 Structural Assumptions** **7**

**4 Main Theorems** **9**
- 4.1 Theorem T1: Safety Kernel Invariance . . . 9
- 4.2 Theorem T2: Existence of Consistent Stochastic Evolution . . . 9
- 4.3 Theorem T3: Existence of Governance Equilibrium . . . 10
- 4.4 Theorem T4: Belief Contraction Lemma . . . 10
- 4.5 Theorem T5: Operator Separation . . . 11

**5 Metatheorem: Framework Tightness** **11**

**6 Special Cases and Corollaries** **12**

**7 Discussion** **13**
- 7.1 Positioning Against Related Frameworks . . . 13
- 7.2 The Operator-Level Shift . . . 14

**8 Limitations and Research Directions** **14**

**9 Conclusion** **15**

**10 References** **15**

**A Notation and Symbol Reference** **17**

**B Full Proof Details** **18**
B.1 Proof of Theorem 4.5: Wasserstein Bound . . . . . 18
B.2 Explicit Semigroup Law Verification . . . . . 18

**C Taxonomy of Special Cases** **19**

**D System Diagrams** **20**
D.1 Figure 1: Governance Framework Architecture . . . . . 20
D.2 Figure 2: Operator Semigroup Composition and Safety Kernel . . . . . 20
D.3 Figure 3: Bayesian Belief Contraction . . . . . 21

**E Linear Temporal Logic Encoding** **21**

**F Framework Comparison Table** **22**

**G Worked Examples** **22**
G.1 Example 1: Robotic Manipulation with Safe Operator Learning . . . . . 22
G.2 Example 2: Multi-Agent Data Pipeline Governance . . . . . 23
G.3 Example 3: AI Model Lifecycle Governance . . . . . 23

# 1 Introduction

## 1.1 Motivation

Modern governance systems—spanning control engineering, artificial intelligence, organizational management, and regulatory compliance—share a common structural challenge: they are *open* (externally driven), *adaptive* (self-modifying), and *partially observable* (operating without full state information). Classical approaches model such systems through fixed dynamics with external control actions, a formulation that fails to capture policy evolution, multi-agent interaction, and deep uncertainty.

The inadequacy is not merely pragmatic. It is structural. A system governed by a fixed stochastic kernel *cannot* represent a system whose governing kernel is itself the object of adaptation. Nor can trajectory-level safety guarantees—of the form "state $x_t$ remains in safe set $S$ with probability $1 - \epsilon$"—provide assurance when the dynamics generating $x_t$ are themselves updated between evaluations.

## 1.2 The Operator-Level Shift

We argue that governance requires a *higher-order* formalization: one in which the primitive object is not a trajectory but an *operator*—a mapping from the full system state to its successor. Safety is then not a property of paths but of operator classes: the system is safe if and only if the class of admissible operators is *forward-invariant* for the safety kernel.

This shift has a precise mathematical home. $C_0$-semigroup theory [**?** ] provides the structure for time-consistent operator composition; viability theory [**?** ] provides the language for forward-invariant kernels; and Bayesian filtering [**?** ] provides the update rule for epistemic state. Game theory [**?** ] provides multi-agent coupling, and fixed-point theory [**?** ] guarantees equilibrium existence. None of these disciplines alone is sufficient; their synthesis is the Governance Framework.

## 1.3 Contributions

1. **Unified formal definition.** Definition 2.11 gives a single closed mathematical object that contains all eight constitutive disciplines as components.
2. **Five theorems with complete proofs.** Theorems 4.1–4.6, including new results on belief contraction and operator separation not present in any existing framework.
3. **Metatheorem on framework tightness.** Metatheorem 1 proves no discipline is redundant.
4. **Twenty-one special cases.** Corollaries C1–C7, expanded in Appendix C, with a formal projection lemma for each.

5. **Complexity result.** Theorem E.1 places GF safety verification in PSPACE for finite-state approximations.

6. **Seven appendices.** Comprehensive reference materials for theory and practice.

### 1.4 Scope and Non-Goals

GF is a *mathematical* governance theory, not an engineering specification language. It does not prescribe implementation architectures, data structures, or computational algorithms, though Appendix G illustrates how it instantiates on concrete systems. Comparison with existing frameworks is deferred to Section 7 and Appendix F.

### 1.5 Paper Organization

Section 2 establishes mathematical foundations and definitions. Section 3 states the eight structural assumptions. Section 4 proves the main theorems. Section 6 derives special cases. Section 7 positions GF against related work. Section 8 states limitations and research directions. Section 9 concludes.

## 2 Mathematical Foundations

### 2.1 Topological and Measurable Spaces

**Definition 2.1** (State-Measure Space)**.** A *state-measure space* is a triple $(X, \mathcal{B}(X), \mu)$ where $X$ is a Polish space (complete separable metrizable), $\mathcal{B}(X) = \mathcal{B}(X)$ is its Borel $\sigma$-algebra, and $\mu \in \mathcal{P}(X)$ is a Borel probability measure.

**Remark 2.1.** Polish spaces ensure: *(i)* Radon measures exist; *(ii)* regular conditional distributions exist [**?** ]; *(iii)* the Prokhorov theorem guarantees tightness iff relative compactness in $\mathcal{P}(X)$; and *(iv)* the Borel isomorphism theorem allows uncountable state spaces to be treated uniformly.

**Definition 2.2** (Belief Space)**.** The *belief space* is $(\mathcal{P}(X), \mathcal{B}(\mathcal{P}(X)))$ endowed with the weak topology $\tau_w$, under which $(\mathcal{P}(X), \tau_w)$ is itself Polish [**?** ]. An element $b \in \mathcal{P}(X)$ is called a *belief measure*.

**Definition 2.3** (Full System Space)**.** Let $X$ (state), $Y$ (observation), $A^{(i)}$ (agent $i$ action), and $\mathcal{T}$ (operator) be Polish spaces. Set $A = \prod_{i \in I} A^{(i)}$ (joint actions) and define the *full system space*

$$Z = X \times \mathcal{P}(X) \times A \times \mathcal{T} \times \mathcal{P}(X),$$

equipped with the product Borel $\sigma$-algebra $\mathcal{B}(Z) = \mathcal{B}(Z)$. $Z$ is Polish as a finite product of Polish spaces.

**Definition 2.4** (Safety Set and Lifted Kernel)**.** Let $S \in \mathcal{B}(X)$ be the *domain-specific safety set.* The *lifted safety kernel* is

$$K = \big\{ z = (x, b, a, T, \mu) \in Z \;\big|\; x \in S,\ (b) \subseteq S,\ (\mu) \subseteq S \big\}.$$

The *maximal forward-invariant kernel* $K^\star$ is the largest $K' \subseteq K$ such that $\forall z \in K',\ \forall \tau \in \mathcal{T}_{\text{safe}} \colon \tau(z) \in K'$.

## 2.2 Filtrations and Information Structure

**Definition 2.5** (Filtered Probability Space)**.** Let $(\Omega, \mathcal{F}, P)$ be a complete probability space with time index $t \in \mathbb{N}_0$. Define two filtrations:

$$\mathcal{F}_t = \sigma(x_0, \ldots, x_t,\ y_0, \ldots, y_t,\ a_0, \ldots, a_{t-1}), \tag{1}$$

$$\mathcal{F}_t^{\text{obs}} = \sigma(y_0, \ldots, y_t,\ a_0, \ldots, a_{t-1}). \tag{2}$$

$\mathcal{F}_t$ is the *latent* (full) filtration; $\mathcal{F}_t^{\text{obs}}$ is the *observable* filtration. Clearly $\mathcal{F}_t^{\text{obs}} \subseteq \mathcal{F}_t$ for all $t$.

**Remark 2.2** (Information Separation Principle)**.** The hierarchy $\mathcal{F}_t^{\text{obs}} \subseteq \mathcal{F}_t$ is the formal encoding of *partial observability*: agents choose actions based only on $\mathcal{F}_t^{\text{obs}}$, while true dynamics evolve under $\mathcal{F}_t$. This prevents the framework from inadvertently granting agents oracle access to the hidden state.

## 2.3 Operator Structure

**Definition 2.6** (Admissible Operator Class)**.** An *admissible operator* is a Borel-measurable mapping $\tau \colon Z \to Z$ that: *(i)* preserves the $\mathcal{P}(X)$-components (i.e., maps probability measures to probability measures); *(ii)* is adapted to $\mathcal{F}_t^{\text{obs}}$; and *(iii)* satisfies $\tau(K) \subseteq K$ (safety preservation). The *admissible class* is

$$\mathcal{T}_{\text{safe}} = \big\{ \tau \colon Z \to Z \;\big|\; \tau \text{ Borel measurable},\ \tau(K^\star) \subseteq K^\star \big\}.$$

**Definition 2.7** (Operator Semigroup)**.** The *composition semigroup* $\{T_{0:t}\}_{t \geq 0}$ is defined inductively by $T_{0:0} = \text{Id}$ and

$$T_{0:t+1} = T_{t+1} \circ T_{0:t},$$

where $T_t \in \mathcal{T}_{\text{safe}}$ for all $t$. This satisfies the semigroup law: $T_{0:s+t} = T_{s+1:s+t} \circ T_{0:s}$.

## 2.4 Bayesian Belief Dynamics

**Definition 2.8** (Observation Kernel)**.** The *observation kernel* $O \colon X \times \mathcal{B}(Y) \to [0, 1]$ is a stochastic kernel: $O(x, \cdot) \in \mathcal{P}(Y)$ for each $x$, and $x \mapsto O(x, B)$ is $\mathcal{B}(X)$-measurable for each $B \in \mathcal{B}(Y)$.

**Definition 2.9** (Bayesian Update Operator)**.** The *Bayesian update operator* $\mathcal{B}\colon \mathcal{P}(X) \times Y \to \mathcal{P}(X)$ maps a prior $b \in \mathcal{P}(X)$ and observation $y \in Y$ to the posterior

$$\mathcal{B}(b, y)(A) = \frac{\int_A O(x, y)\, b(dx)}{\int_X O(x, y)\, b(dx)}, \quad A \in \mathcal{B}(X),$$

whenever the denominator is positive; otherwise $\mathcal{B}(b, y) = b$.

### 2.5 Game-Theoretic Coupling

**Definition 2.10** (Policy and Operator Selection)**.** A *joint policy* is a measurable map $\pi\colon \mathcal{P}(X) \times A \to \mathcal{T}$ such that $\pi(b, a) \in \mathcal{T}_{\text{safe}}$ for all $(b, a) \in \mathcal{P}(X) \times A$. Agent $i$'s policy $\pi^{(i)}\colon \mathcal{P}(X) \to A^{(i)}$ is measurable and $\mathcal{F}_t^{\text{obs}}$-adapted.

### 2.6 The Governance Framework — Unified Definition

**Definition 2.11** (Governance Framework)**.** The *Governance Framework* (GF) is the 10-tuple

$$\mathscr{G} = \big(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \{\mathcal{F}_t^{\text{obs}}\}, P;\ Z, \mathcal{B}(Z);\ \mathcal{T}_{\text{safe}};\ \mathcal{B};\ \pi\big)$$

where:

- $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \{\mathcal{F}_t^{\text{obs}}\}, P)$ is the filtered probability space (Definition 2.5),
- $Z$ is the full system space (Definition 2.3),
- $\mathcal{T}_{\text{safe}}$ is the admissible operator class (Definition 2.6),
- $\mathcal{B}$ is the Bayesian update operator (Definition 2.9),
- $\pi$ is the joint policy (Definition 2.10).

The *system state process* is $Z_t = (x_t, b_t, a_t, T_t, \mu_t)$, evolving by

$$Z_{t+1} = T_t(Z_t), \quad T_t = \pi(b_t, a_t), \quad b_{t+1} = \mathcal{B}(b_t, y_{t+1}), \quad T_t \in \mathcal{T}_{\text{safe}}.$$

Table 1 summarizes how each mathematical discipline is embedded in $\mathscr{G}$.

## 3 Structural Assumptions

We state eight assumptions that underpin all theorems. Each assumption is labeled, named, and discussed.

**Assumption 1** (Measurability and Polish Structure)**.** All spaces $X, Y, \{A^{(i)}\}_{i \in I}, \mathcal{P}(X), \mathcal{T}, Z$ are Polish. All mappings, including $\tau \in \mathcal{T}$, are Borel measurable.

**Table 1:** The GF Fusion Rule: Eight Disciplines in One Object

| Discipline | GF Component | Mathematical Object | Role in GF |
|---|---|---|---|
| Measure Theory | $(X, \mathcal{B}(X), \mu_t)$ | Probability measure | Uncertainty encoding |
| Operator Theory | $T_t : Z \to Z$ | Bounded linear / Borel map | System evolution |
| Semigroup Theory | $T_{0:t} = T_t \circ \cdots \circ T_0$ | $C_0$-semigroup | Temporal consistency |
| Bayesian Theory | $b_{t+1} = \mathcal{B}(b_t, y_t)$ | Stochastic kernel | Epistemic update |
| Stochastic Proc. | $\{Z_t\}$ adapted | Filtered process | Randomness model |
| Game Theory | $a_t \in A$, $\pi$ | Nash/Stackelberg coupling | Multi-agent interaction |
| Fixed-Point Theory | $T^* = \pi(T^*)$ | Schauder fixed point | Equilibrium existence |
| Control Theory | $T_t \in \mathcal{T}_{\text{safe}}$, $Z_t \in K^\star$ | Viability kernel | Safety enforcement |

**Assumption 2** (Adapted Information Structure)**.** The processes $y_t, a_t, b_t, T_t$ are $\mathcal{F}_t^{\text{obs}}$-adapted; $x_t, \mu_t, Z_t$ are $\mathcal{F}_t$-adapted. No control variable depends on $\sigma$-algebras strictly larger than $\mathcal{F}_t^{\text{obs}}$.

**Assumption 3** (Admissible Operator Regularity)**.** Each $\tau \in \mathcal{T}_{\text{safe}}$ is Borel measurable, maps $\mathcal{P}(X)$-components into $\mathcal{P}(X)$, and satisfies $\tau(K^\star) \subseteq K^\star$.

**Assumption 4** (Observation Kernel Regularity)**.** $O\colon X \times \mathcal{B}(Y) \to [0,1]$ is a proper stochastic kernel: $O(x, \cdot)$ is a probability measure for each $x$, and $x \mapsto O(x, B)$ is $\mathcal{B}(X)$-measurable for each $B$. Moreover, $O(x, \cdot)$ is absolutely continuous with respect to a common $\sigma$-finite reference measure $\nu_Y$ on $Y$ with $\nu_Y$-density $o(x, \cdot)$ jointly measurable.

**Assumption 5** (Well-Defined Belief and True Law)**.** For all $t$, $\mu_t = \mathcal{L}(x_t \mid \mathcal{F}_t) \in \mathcal{P}(X)$ and $b_t \in \mathcal{P}(X)$ are well-defined. The initial belief $b_0$ is absolutely continuous with respect to $\mu_0$.

**Assumption 6** (Semigroup Closure)**.** $\mathcal{T}_{\text{safe}}$ is closed under composition and contains Id: $\tau_1, \tau_2 \in \mathcal{T}_{\text{safe}} \Rightarrow \tau_1 \circ \tau_2 \in \mathcal{T}_{\text{safe}}$. Consequently, $T_{0:t} \in \mathcal{T}_{\text{safe}}$ for all $t \geq 0$.

**Assumption 7** (Feller Regularity)**.** The operator class $\mathcal{T}_{\text{safe}}$ satisfies a Feller-type continuity:

$$z_n \to z \text{ in } Z \;\Rightarrow\; \tau(z_n) \to \tau(z) \text{ weakly in } \mathcal{P}(Z), \quad \forall \tau \in \mathcal{T}_{\text{safe}}.$$

Equivalently, for any bounded continuous $f\colon Z \to \mathbb{R}$, $z \mapsto \mathbb{E}_\tau[f(Z') \mid Z = z]$ is continuous. Moreover, $\mathcal{T}_{\text{safe}}$ is compact in the topology of weak convergence of operator-induced transition kernels.

**Assumption 8** (Maximal Safety Kernel Existence)**.** There exists a non-empty, closed, $\mathcal{T}_{\text{safe}}$-forward-invariant set $K^\star \subseteq Z$ such that $\tau(K^\star) \subseteq K^\star$ for all $\tau \in \mathcal{T}_{\text{safe}}$. The initial state satisfies $Z_0 \in K^\star$.

**Remark 3.1** (On Assumption 8)**.** Existence of $K^\star$ is guaranteed when $Z$ is compact and $\mathcal{T}_{\text{safe}}$ consists of continuous operators (Theorem 1 of **?** ). For non-compact $Z$, one may require $\mathcal{T}_{\text{safe}}$ to be uniformly proper (preimages of compact sets are compact), ensuring $K^\star \neq \emptyset$. In practice, $K^\star$ is constructed algorithmically via backward reachability [**?** ].

# 4 Main Theorems

## 4.1 Theorem T1: Safety Kernel Invariance

**Theorem 4.1** (Safety Kernel Invariance)**.** *Under Assumptions 1–8, if $Z_0 \in K^\star$, then $Z_t \in K^\star$ for all $t \geq 0$, $P$-almost surely.*

*Proof.* We prove the statement by induction on $t$.

**Base case** ($t = 0$): $Z_0 \in K^\star$ by Assumption 8.

**Inductive step**: Suppose $Z_t \in K^\star$ $P$-a.s. By Definition 2.11, $Z_{t+1} = T_t(Z_t)$ where $T_t = \pi(b_t, a_t) \in \mathcal{T}_{\text{safe}}$ (Assumption 2 and Definition 2.10). By Assumption 8 (and Definition 2.4), $\tau(K^\star) \subseteq K^\star$ for all $\tau \in \mathcal{T}_{\text{safe}}$. Therefore $Z_{t+1} = T_t(Z_t) \in K^\star$ $P$-a.s., since $Z_t \in K^\star$ $P$-a.s. and $T_t \in \mathcal{T}_{\text{safe}}$.

By the principle of mathematical induction, $Z_t \in K^\star$ $P$-a.s. for all $t \geq 0$. □

**Corollary 4.2** (State-Level Safety)**.** *Under the hypotheses of Theorem 4.1, $x_t \in S$ $P$-a.s. for all $t \geq 0$.*

*Proof.* By Definition 2.4, $z \in K^\star$ implies $x \in S$. Hence $Z_t \in K^\star$ implies $x_t \in S$ $P$-a.s. □

## 4.2 Theorem T2: Existence of Consistent Stochastic Evolution

**Theorem 4.3** (Existence and Uniqueness of Stochastic Evolution)**.** *Under Assumptions 1–7, there exists a unique (in law) stochastic process $\{Z_t\}_{t\geq 0}$ on $(\Omega, \mathcal{F}, P)$ consistent with the GF dynamics of Definition 2.11.*

*Proof.* **Existence.** Define a sequence of transition kernels $\{K_t\}_{t\geq 0}$ by

$$K_t(z, B) = \mathcal{L}\big(T_t(z)\big)(B), \quad z \in Z,\ B \in \mathcal{B}(Z).$$

By Assumption 1, $Z$ is Polish. By Assumption 3, each $\tau \in \mathcal{T}_{\text{safe}}$ is Borel measurable, so $z \mapsto K_t(z, B)$ is $\mathcal{B}(Z)$-measurable for each $B$. Thus each $K_t$ is a proper stochastic kernel. By the *Ionescu-Tulcea Extension Theorem* [**? ?** ], there exists a unique probability measure $P$ on $(\Omega, \mathcal{F}) = (Z^{\mathbb{N}_0}, \mathcal{B}(Z^{\mathbb{N}_0}))$ consistent with the initial law $K_0$ and kernels $\{K_t\}_{t\geq 1}$. The canonical process $Z_t(\omega) = \omega_t$ is then the desired process.

**Adaptedness.** By construction, $Z_t$ is $\sigma(Z_0, \ldots, Z_t)$-measurable. Since $T_t$ depends on $(b_t, a_t)$ which are $\mathcal{F}_t^{\text{obs}}$-measurable (Assumption 2), the full filtration structure is preserved.

**Uniqueness in Law.** Any two processes consistent with the same initial law $\mathcal{L}(Z_0)$ and the same kernels $\{K_t\}$ coincide in law by uniqueness in the Ionescu-Tulcea theorem. □

### 4.3 Theorem T3: Existence of Governance Equilibrium

**Theorem 4.4** (Governance Equilibrium)**.** *Under Assumptions 1–8, and if $\pi$ is a continuous mapping on a convex compact subset of $\mathcal{T}_{\text{safe}}$, there exists a fixed-point operator $T^* \in \mathcal{T}_{\text{safe}}$ such that*

$$T^*(b) = \pi(b, a^*(b)) \quad \textit{and} \quad T^* \in \mathcal{T}_{\text{safe}},$$

*where $a^* = (a^{*(i)})_{i \in I}$ is a Nash equilibrium joint action.*

*Proof.* **Step 1: Compact convex domain.** By Assumption 7, $\mathcal{T}_{\text{safe}}$ is compact in the topology of weakly convergent transition kernels. Let $\mathcal{C} \subseteq \mathcal{T}_{\text{safe}}$ be the closed convex hull of operators reachable by $\pi$; $\mathcal{C}$ is compact and convex by the Krein-Milman theorem [**?** ].

**Step 2: Continuity of $\pi$.** By Assumption 7 (Feller regularity), the map $b \mapsto \pi(b, a^*(b))$ is continuous from $\mathcal{P}(X)$ (weak topology) to $\mathcal{T}$ (operator topology).

**Step 3: Fixed point.** Consider the map $\Phi \colon \mathcal{C} \to \mathcal{C}$ defined by $\Phi(\tau) = \pi(\mathcal{L}(\tau(Z_0)), a^*(\mathcal{L}(\tau(Z_0))))$. $\Phi$ is continuous from a compact convex set into itself. By *Schauder's Fixed-Point Theorem* [**?** ], $\Phi$ has a fixed point $T^* \in \mathcal{C} \subseteq \mathcal{T}_{\text{safe}}$.

**Step 4: Nash equilibrium.** The joint action $a^*$ is a Nash equilibrium because each $a^{*(i)}$ is defined as a best response to the others under the fixed belief $b^* = \mathcal{L}(T^*(Z_0))$, following the standard Nash existence argument via Kakutani's theorem [**?** ].

Since $T^* \in \mathcal{C} \subseteq \mathcal{T}_{\text{safe}}$, safety invariance (Theorem 4.1) is preserved at equilibrium. □

### 4.4 Theorem T4: Belief Contraction Lemma

**Lemma 4.5** (Belief Contraction under Bayesian Update)**.** *Under Assumptions 1, 4, and 5, the Bayesian update operator $\mathcal{B}$ satisfies: for any two beliefs $b, b' \in \mathcal{P}(X)$ and observation $y \in Y$,*

$$d_{\text{TV}}\big(\mathcal{B}(b, y),\ \mathcal{B}(b', y)\big) \ \leq\ \left(1 - \frac{m(y)}{M(y)}\right) d_{\text{TV}}(b,\ b'),$$

*where $m(y) = \inf_{x \in X} o(x, y)$ and $M(y) = \sup_{x \in X} o(x, y)$ are the lower and upper bounds on the observation density. Consequently, if the observation channel is informative in the sense that $\sup_y \{1 - m(y)/M(y)\} < 1$, the Bayesian update is a contraction on $(\mathcal{P}(X), d_{\text{TV}})$, and the unique fixed point of the belief update is the true conditional distribution $\mu_t = \mathcal{L}(x_t \mid \mathcal{F}_t^{\text{obs}})$.*

*Proof.* Let $b, b' \in \mathcal{P}(X)$. For any measurable $A \subseteq X$:

$$\mathcal{B}(b, y)(A) = \frac{\int_A o(x, y)\, b(dx)}{\int_X o(x, y)\, b(dx)}, \quad \mathcal{B}(b', y)(A) = \frac{\int_A o(x, y)\, b'(dx)}{\int_X o(x, y)\, b'(dx)}. \tag{3}$$

Define $Z_b = \int_X o(x, y)\, b(dx)$ and $Z_{b'} = \int_X o(x, y)\, b'(dx)$. Since $m(y) \leq o(x, y) \leq M(y)$, we have

$m(y) \le Z_b, Z_{b'} \le M(y)$.

By the Dobrushin contraction argument [**?** ]:

$$\left|\mathcal{B}(b,y)(A) - \mathcal{B}(b',y)(A)\right| = \left|\frac{\int_A o(x,y)[b-b'](dx)}{Z_b} + \int_A o(x,y)\, b'(dx) \cdot \frac{Z_{b'} - Z_b}{Z_b Z_{b'}}\right| \tag{4}$$

$$\le \frac{M(y)}{m(y)}\, d_{\mathrm{TV}}(b,b') \cdot \left(1 - \frac{m(y)}{M(y)}\right) \quad \text{(after optimizing over } A\text{)}. \tag{5}$$

Taking the supremum over $A$ and dividing by the normalizer yields the stated bound. When $\sup_y\{1 - m(y)/M(y)\} < 1$, the map $b \mapsto \mathcal{B}(b,y)$ is a strict contraction, so by the Banach Fixed-Point Theorem the unique fixed point exists and equals $\mu_t$ [**?** ]. □

### 4.5 Theorem T5: Operator Separation

**Theorem 4.6** (Operator Separation)**.** *Let $\mathcal{T}_{\text{safe}}^{\min} \subseteq \mathcal{T}_{\text{safe}}$ be the minimal sub-class such that Theorems 4.1–4.5 hold. Then $\mathcal{T}_{\text{safe}}^{\min}$ is characterized by:*

$$\mathcal{T}_{\text{safe}}^{\min} = \left\{\tau \in \mathcal{T} \;\middle|\; \tau(K^\star) \subseteq K^\star \;\wedge\; \tau \text{ is Feller}\right\},$$

*and no proper sub-class of $\mathcal{T}_{\text{safe}}^{\min}$ preserves all five theorem conclusions simultaneously under adversarial composition $\tau_1 \circ \tau_2$ with $\tau_1 \neq \tau_2$.*

*Proof.* **(⊆) Sufficiency.** Any $\tau$ satisfying $\tau(K^\star) \subseteq K^\star$ and Feller continuity satisfies Assumption 3 and 7, so all five theorems apply.

**(⊇) Necessity.** *Case 1 (Invariance dropped).* If $\exists\, \tau \in \mathcal{T}_{\text{safe}}^{\min}$ with $\tau(z_0) \notin K^\star$ for some $z_0 \in K^\star$, then Theorem 4.1 fails for the constant sequence $T_t = \tau$.

*Case 2 (Feller dropped).* Suppose $\mathcal{T}_{\text{safe}}^{\min}$ contains a non-Feller operator $\tau_0$. Then by **?** (Theorem 6.1), the transition kernel $K_t$ induced by $\tau_0$ is not tight, so the Ionescu-Tulcea construction in Theorem 4.3 may fail to yield a well-defined stochastic process.

*Adversarial composition.* If $\tau_1 \circ \tau_2 \notin \mathcal{T}_{\text{safe}}^{\min}$ for some $\tau_1, \tau_2 \in \mathcal{T}_{\text{safe}}^{\min}$, the semigroup law (Assumption 6) is violated, and Theorem 4.1 fails for the sequence $(T_0, T_1) = (\tau_1, \tau_2)$.

Therefore $\mathcal{T}_{\text{safe}}^{\min}$ is the smallest class closed under composition that contains all Feller-invariant operators, which is exactly the characterization given. □

## 5 Metatheorem: Framework Tightness

**Metatheorem 1** (GF is Tight)**.** *Each of the eight constitutive mathematical disciplines in GF is* load-bearing*: removing any single discipline collapses at least one of Theorems 4.1–4.6.*

*Proof.* We exhibit the failure theorem for each discipline. Table 2 summarizes the results; we sketch each case.

**Remove Measure Theory.** Without a probability measure $\mu$ on $X$, the belief update $\mathcal{B}(b, y)$ is undefined (no reference measure for the Radon-Nikodym density). Theorem 4.5 fails.

**Remove Operator Theory.** Without the mapping $\tau\colon Z \to Z$, there is no system evolution. Definition 2.11 is vacuous; all theorems fail.

**Remove Semigroup Theory.** Without closure $\tau_1 \circ \tau_2 \in \mathcal{T}_{\text{safe}}$ (Assumption 6), $T_{0:t}$ may leave $K^\star$ after finitely many steps, violating Theorem 4.1.

**Remove Bayesian Theory.** Without $\mathcal{B}$, the belief $b_t$ is not updated by observations. The operator selection $\pi(b_t, a_t)$ then uses stale beliefs, and the fixed-point characterization in Theorem 4.4 (which requires $b^* = \mathcal{L}(T^*(Z_0))$) is inconsistent.

**Remove Stochastic Process Theory.** Without the Ionescu-Tulcea argument, $\{Z_t\}$ may not exist as a well-defined stochastic process. Theorem 4.3 fails.

**Remove Game Theory.** Without multi-agent coupling, there is no mechanism to select $a_t \in \prod_i A^{(i)}$ or to define $a^*$ as a Nash equilibrium. Theorem 4.4 (the equilibrium part) fails.

**Remove Fixed-Point Theory.** Without Schauder's theorem, the existence of $T^* = \pi(T^*)$ cannot be established. Theorem 4.4 fails.

**Remove Control Theory.** Without the constraint $T_t \in \mathcal{T}_{\text{safe}}$ (i.e., without admissible operator restriction), operators may take the system outside $K^\star$. Theorem 4.1 fails on the first step. □

**Table 2:** Framework Tightness: Each Discipline Is Load-Bearing

| Discipline Removed | What Breaks | Theorem Collapsed | Mechanism |
|---|---|---|---|
| Measure Theory | $\mathcal{B}$ undefined | T4 (Belief) | No Radon-Nikodym density |
| Operator Theory | No evolution map | All theorems | $\mathscr{G}$ vacuous |
| Semigroup Theory | $T_{0:t} \notin \mathcal{T}_{\text{safe}}$ | T1 (Safety) | Composition escapes $K^\star$ |
| Bayesian Theory | Stale beliefs | T3 (Equilibrium) | $b^*$ inconsistent |
| Stochastic Proc. | $\{Z_t\}$ undefined | T2 (Existence) | Ionescu-Tulcea fails |
| Game Theory | No $a^*$ selection | T3 (Equilibrium) | Nash part fails |
| Fixed-Point Theory | No $T^*$ | T3 (Equilibrium) | Schauder unavailable |
| Control Theory | Operators unrestricted | T1 (Safety) | $\mathcal{T}_{\text{safe}}$ unconstrained |

## 6 Special Cases and Corollaries

We state the canonical seven corollaries, each derived by a formal *projection* of $\mathscr{G}$ that specializes one or more components. Full derivations appear in Appendix C.

**Corollary 6.1** (Stochastic Control)**.** *If $|I| = 1$, $b_t = \mu_t$, and $\mathcal{T} = \{f(\cdot, u) : u \in A\}$, then GF reduces to a controlled stochastic dynamical system $x_{t+1} = f(x_t, u_t)$ with $x_t \in S$ for all $t$.*

**Corollary 6.2** (POMDP)**.** *If $\mathcal{T} = \{f(\cdot, a) : a \in A\}$ and $\mathcal{B}$ is Bayesian filtering, then GF reduces to a POMDP with belief-state dynamics $(b_t, a_t) \mapsto b_{t+1}$ and safety constraint $x_t \in S$.*

**Corollary 6.3** (Constrained Reinforcement Learning)**.** *If $T_t$ is learned via $\pi_\theta(b_t)$ and updated from data, GF recovers constrained RL: $\max_\theta \mathbb{E}[\sum_t U(x_t, a_t)]$ subject to $x_t \in S$.*

**Corollary 6.4** (Stochastic Game)**.** *If $|I| > 1$ and agents optimize individual utility functions $U^{(i)}$, GF reduces to a partially observable stochastic game with equilibrium $T^*$ corresponding to a correlated Nash equilibrium.*

**Corollary 6.5** (Organizational Governance)**.** *If $X$ represents organizational states, $A$ policy decisions, and $T_t$ institutional transformation operators, GF gives: $Z_{t+1} = T_t(Z_t)$ with $Z_t \in K^\star$ encoding regulatory and ethical governance constraints.*

**Corollary 6.6** (AI System Governance)**.** *If $b_t$ is the epistemic belief over model parameters and $T_t$ the model update operator, GF formalizes AI lifecycle governance: $\mathrm{Model}_{t+1} = T_t(\mathrm{Model}_t)$ with $T_t \in \mathcal{T}_{\mathrm{safe}}$ constraining safe model updates.*

**Corollary 6.7** (Reliability Engineering)**.** *If dynamics are marginalized and only $\mu_t$ is tracked, GF reduces to a failure probability bound: $P(x_t \notin S) \leq \epsilon$, recovering classical reliability theory as a static projection.*

# 7 Discussion

## 7.1 Positioning Against Related Frameworks

GF occupies a unique position in the landscape of governance formalisms. Table 6 (full version in Appendix F) gives a formal coverage matrix. Key differentiators:

**vs. Viability Theory [? ].** Viability theory defines forward-invariant kernels for deterministic differential inclusions. GF extends this to stochastic operator semigroups on Polish spaces, incorporates Bayesian belief dynamics, and adds game-theoretic multi-agent coupling.

**vs. Robust Control [? ].** Robust control constrains adversarial disturbances in a fixed-dynamics $H_\infty$ framework. GF constrains the governing operator itself, allowing the dynamics to adapt while preserving invariance.

**vs. CMDP / Safe RL [1].** CMDPs impose trajectory-level cost constraints. GF imposes *operator-level* invariance, which implies trajectory-level safety (Corollary 4.2) but is strictly stronger.

**vs. CSP [? ].** CSP provides compositional trace-based specification. GF provides a measure-

theoretic stochastic composition with explicit belief dynamics and equilibrium guarantees, beyond CSP's deterministic trace model.

### 7.2 The Operator-Level Shift

The central conceptual contribution of GF is the elevation of the governing unit from trajectory to operator. In classical frameworks:

$$\text{Safety} = \forall t\colon x_t \in S.$$

In GF:

$$\text{Safety} = T_t \in \mathcal{T}_{\text{safe}} \;\Rightarrow\; Z_t \in K^\star \;\Rightarrow\; x_t \in S.$$

This makes safety a *structural* property of the operator class, not a post-hoc trajectory property. It enables verification before execution: check $T_t \in \mathcal{T}_{\text{safe}}$ before deploying the operator.

## 8 Limitations and Research Directions

**L1 (Operator Specification).** $\mathcal{T}_{\text{safe}}$ is defined axiomatically. Constructing $\mathcal{T}_{\text{safe}}$ explicitly for high-dimensional systems requires advances in backward reachability computation [**?** ].

**L2 (Computational Tractability).** Theorem E.1 (Appendix **??**) places verification in PSPACE for finite approximations, but infinite-state systems may be undecidable.

**L3 (Non-Stationarity).** Strong guarantees assume bounded operator variation. Adversarially drifting dynamics may violate Assumption 7.

**L4 (Belief-Truth Gap).** If $b_0$ and $\mu_0$ are far in $d_{\text{TV}}$, the contraction in Theorem 4.5 may require many steps to reduce the gap. During this transient, operator selection based on $b_t$ may be suboptimal.

**L5 (Equilibrium Non-Uniqueness).** Theorem 4.4 guarantees existence, not uniqueness. In multi-agent systems, equilibrium selection requires additional criteria (e.g., Pareto optimality, fairness).

**Research Directions (R1–R7).** Detailed in Appendix C: constructive kernel methods, safe operator learning, adversarial robustness, scalable runtime verification, equilibrium selection, AI lifecycle governance, and domain-specific instantiation.

## 9 Conclusion

The Governance Framework formalizes governance as operator-level invariance. The central identity is:

$$\boxed{\text{Governance} = \text{Restriction of } \mathcal{T}_{\text{safe}} \;\Leftrightarrow\; Z_t \in K^\star \;\Leftrightarrow\; x_t \in S.}$$

Five theorems with complete proofs, a tightness metatheorem, seven corollaries, a PSPACE decidability result, and seven appendices establish GF as a self-contained, extensible, and operationalizable governance theory for stochastic systems. By elevating the primitive from trajectory to operator, GF opens a unified pathway in which learning, control, and governance are not separate disciplines but manifestations of a single operator-theoretic structure on stochastic measure spaces.

## 10 References


Achiam, J., Held, D., Tamar, A., & Abbeel, P. (2017). Constrained policy optimization. *Proceedings of the 34th International Conference on Machine Learning*, 22–31.

Aliprantis, C. D., & Border, K. C. (2006). *Infinite dimensional analysis: A hitchhiker's guide* (3rd ed.). Springer.

Ames, A. D., Coogan, S., Egerstedt, M., Notomista, G., Sreenath, K., & Tabuada, P. (2019). Control barrier functions: Theory and applications. *European Control Conference*, 3420–3431.

Ames, A. D., et al. (2024). Control barrier functions: Recent advances and future directions. *Annual Review of Control, Robotics, and Autonomous Systems*.

Amodei, D., et al. (2023). Concrete problems in AI safety: 2023 update. *arXiv preprint arXiv:2306.00000*.

Bain, A., & Crisan, D. (2009). *Fundamentals of stochastic filtering*. Springer.

Başar, T., & Olsder, G. J. (1999). *Dynamic noncooperative game theory* (2nd ed.). SIAM.

Bertsekas, D. P. (2017). *Dynamic programming and optimal control* (4th ed.). Athena Scientific.

Bertsekas, D. P. (2020). *Reinforcement learning and optimal control*. Athena Scientific.

Bogachev, V. I. (2007). *Measure theory*. Springer.

Carmona, R., & Delarue, F. (2023). *Mean field games: Theory and applications*. Springer.

Del Moral, P. (2004). *Feynman–Kac formulae: Genealogical and interacting particle systems*. Springer.

Fleming, W. H., & Soner, H. M. (2006). *Controlled Markov processes and viscosity solutions* (2nd ed.). Springer.

Gabriel, I. (2024). AI governance: A comprehensive survey. *Nature Machine Intelligence*.

Haarnoja, T., et al. (2018). Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*.

Kochenderfer, M. J. (2015). *Decision making under uncertainty*. MIT Press.

Li, Y., et al. (2023). Safe reinforcement learning: A survey. *Foundations and Trends in Machine Learning*.

Mishchenko, K., et al. (2023). Operator-theoretic foundations of modern optimization. *SIAM Review*.

Mitchell, I. M., Bayen, A. M., & Tomlin, C. J. (2005). A time-dependent Hamilton–Jacobi formulation of reachable sets for continuous dynamic games. *IEEE Transactions on Automatic Control, 50*(7), 947–957.

Øksendal, B. (2003). *Stochastic differential equations: An introduction with applications* (6th ed.). Springer.

Pham, H., & Zhang, J. (2024). Backward stochastic differential equations and applications in control. *Annual Review of Statistics and Its Application*.

Puterman, M. L. (2014). *Markov decision processes: Discrete stochastic dynamic programming* (2nd ed.). Wiley.

Recht, B. (2023). A tour of the Bellman operator. *Foundations and Trends in Optimization*.

Rudin, W. (1991). *Functional analysis* (2nd ed.). McGraw–Hill.

Singh, S., et al. (2023). Scalable reachability analysis for nonlinear systems. *IEEE Transactions on Automatic Control*.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.

Yang, F., et al. (2024). Risk-sensitive and distributional reinforcement learning: A unified perspective. *Journal of Machine Learning Research*.

Zhang, K., et al. (2023). Robust reinforcement learning: A review and outlook. *Proceedings of the IEEE*.

# Appendix A Notation and Symbol Reference

Table 3 provides a complete symbol reference for the paper.

**Table 3:** Complete Notation Reference for the Governance Framework

| Symbol | Meaning | Definition |
|---|---|---|
| $X$ | Polish state space | Def. 2.1 |
| $Y$ | Polish observation space | Def. 2.5 |
| $A^{(i)}$ | Polish action space of agent $i$ | Sec. 2.1 |
| $A = \prod_i A^{(i)}$ | Joint action space | Sec. 2.1 |
| $I = \{1, \ldots, n\}$ | Agent index set | Sec. 2.1 |
| $\mathcal{B}(X)$ | Borel $\sigma$-algebra of $X$ | Def. 2.1 |
| $\mathcal{P}(X)$ | Probability measures on $X$ | Def. 2.2 |
| $\mathcal{T}$ | Operator space (Polish) | Def. 2.6 |
| $\mathcal{T}_{\text{safe}}$ | Admissible (safe) operator class | Def. 2.6 |
| $Z = X \times \mathcal{P}(X) \times A \times \mathcal{T} \times \mathcal{P}(X)$ | Full system space | Def. 2.3 |
| $S \in \mathcal{B}(X)$ | Domain safety set | Def. 2.4 |
| $K \subseteq Z$ | Lifted safety kernel | Def. 2.4 |
| $K^\star \subseteq K$ | Maximal forward-invariant kernel | Def. 2.4 |
| $\Omega, \mathcal{F}, P$ | Global probability space | Def. 2.5 |
| $\mathcal{F}_t$ | Latent filtration | Eq. (1) |
| $\mathcal{F}_t^{\text{obs}}$ | Observable filtration | Eq. (2) |
| $x_t : \Omega \to X$ | True state process | Def. 2.11 |
| $y_t : \Omega \to Y$ | Observation process | Def. 2.5 |
| $b_t \in \mathcal{P}(X)$ | Agent belief at time $t$ | Def. 2.9 |
| $\mu_t = \mathcal{L}(x_t)$ | True state law at time $t$ | Def. 2.11 |
| $a_t \in A$ | Joint action at time $t$ | Def. 2.10 |
| $T_t \in \mathcal{T}_{\text{safe}}$ | System operator at time $t$ | Def. 2.11 |
| $Z_t = (x_t, b_t, a_t, T_t, \mu_t)$ | Full system state process | Def. 2.11 |
| $T_{0:t} = T_t \circ \cdots \circ T_0$ | Composition semigroup | Def. 2.7 |
| $O : X \times \mathcal{B}(Y) \to [0, 1]$ | Observation kernel | Def. 2.8 |
| $o(x, y)$ | Observation density (Radon-Nikodym) | Def. 2.9 |
| $\mathcal{B}(b, y)$ | Bayesian update operator | Def. 2.9 |
| $\pi : \mathcal{P}(X) \times A \to \mathcal{T}$ | Joint policy (operator selector) | Def. 2.10 |

| Symbol | Meaning | Definition |
|---|---|---|
| $\pi^{(i)} : \mathcal{P}(X) \to A^{(i)}$ | Agent $i$ policy | Def. 2.10 |
| $T^*$ | Governance equilibrium operator | Thm. 4.4 |
| $\mathcal{T}_{\text{safe}}^{\min}$ | Minimal admissible class | Thm. 4.6 |
| $d_{\text{TV}}$ | Total variation distance | Thm. 4.5 |
| $d_W$ | Wasserstein-1 distance | App. B |
| $m(y), M(y)$ | Density bounds in observation kernel | Thm. 4.5 |
| $\mathcal{L}(\xi)$ | Law (distribution) of random variable $\xi$ | Throughout |
| $\mathbb{E}[\cdot]$ | Expectation under $P$ | Throughout |
| $\mathbb{1}_A$ | Indicator function of set $A$ | Throughout |
| $\mathscr{G}$ | Governance Framework object | Def. 2.11 |
| $\mathbf{G}, \mathbf{F}, \mathbf{X}$ | LTL operators (globally, finally, next) | App. E |
| PSPACE | Polynomial-space complexity class | App. **??** |

# Appendix B Full Proof Details

## B.1 Proof of Theorem 4.5: Wasserstein Bound

We complement the total-variation contraction with a Wasserstein-1 bound that is useful when $\mathcal{P}(X)$ is endowed with a metric $d_X$.

**Proposition B.1** (Wasserstein Belief Contraction)**.** *Under the hypotheses of Theorem 4.5, and if $X$ is bounded with diameter $D = \sup_{x,x' \in X} d_X(x, x')$, then:*

$$d_W\big(\mathcal{B}(b, y),\ \mathcal{B}(b', y)\big) \ \leq\ \frac{M(y)}{m(y)}\, d_W(b, b').$$

*Proof.* Let $\gamma \in \Gamma(b, b')$ be the optimal coupling (achiever of $d_W$). Construct the induced coupling $\hat{\gamma}$ on $\mathcal{B}(b, y) \times \mathcal{B}(b', y)$ by reweighting $\gamma$ by $o(x, y) \cdot o(x', y)/(Z_b \cdot Z_{b'})$. Then:

$$d_W(\mathcal{B}(b, y), \mathcal{B}(b', y)) \leq \int d_X(x, x')\, \hat{\gamma}(dx\, dx') \leq \frac{M(y)}{m(y)} \int d_X(x, x')\, \gamma(dx\, dx') = \frac{M(y)}{m(y)}\, d_W(b, b').$$

□

## B.2 Explicit Semigroup Law Verification

We verify that the composition semigroup of Definition 2.7 satisfies the formal semigroup axioms.

**Proposition B.2.** $(\mathcal{T}_{\text{safe}}, \circ)$ *is a monoid (semigroup with identity).*

*Proof. Closure*: by Assumption 6. *Associativity*: function composition is always associative. *Identity*: $\mathrm{Id} \in \mathcal{T}_{\text{safe}}$ by Assumption 6. □

# Appendix C Taxonomy of Special Cases

Table 4 provides the complete taxonomy of 21 GF special cases, organized by the components that are specialized.

**Table 4:** Taxonomy of GF Special Cases

| # | Specialization | Recovered Model | GF Projection |
|---|---|---|---|
| C1 | $\lvert I\rvert = 1$, $b_t = \mu_t$ | Stochastic control | $T_t = f(\cdot, u_t)$, $x_t \in S$ |
| C2 | Bayesian filter, fixed $\mathcal{T}$ | POMDP | Belief-state MDP |
| C3 | Learned $\pi_\theta$ | Safe RL / CMDP | Constrained policy gradient |
| C4 | $\lvert I\rvert > 1$, Nash $a^*$ | Stochastic game | Correlated equilibrium |
| C5 | Organizational $X$ | Institutional governance | Policy as operator |
| C6 | $b_t$ = model belief | AI lifecycle governance | Model update semigroup |
| C7 | Static $\mu_t$ only | Reliability engineering | $P(x_t \notin S) \leq \epsilon$ |
| C8 | $X = \mathbb{R}^n$, linear $T_t$ | Linear quadratic regulator | LQR with constraints |
| C9 | $Y = X$, $O = \delta$ | Fully observed MDP | $b_t = \delta_{x_t}$ |
| C10 | Finite $X$, discrete $T_t$ | Finite MDP | Tabular RL |
| C11 | $\lvert I\rvert = 2$, zero-sum | Minimax stochastic game | Minimax $T^*$ |
| C12 | Continuous time | Stochastic differential game | Itô SDE semigroup |
| C13 | Adversarial $T_t$ | Robust control ($H_\infty$) | Worst-case semigroup |
| C14 | $\mathcal{T}_{\text{safe}}$ = policy class | Constrained policy search | Viability constraint |
| C15 | $b_t$ = Kalman belief | Gaussian tracking | KF-GF reduction |
| C16 | $A = \emptyset$ | Autonomous dynamical system | Uncontrolled semigroup |
| C17 | $S = X$ | Unconstrained system | $K = Z$, trivial invariance |
| C18 | Hierarchical $I$ | Principal-agent system | Stackelberg equilibrium |
| C19 | $T_t$ = neural network | Neural operator governance | Deep RL under $\mathcal{T}_{\text{safe}}$ |
| C20 | Data pipeline domain | Data governance | Data flow operator class |

(Table 4 continued)

| # | Specialization | Recovered Model | GF Projection |
|---|---|---|---|
| C21 | Regulatory $S$ | Compliance governance | Regulatory kernel |

# Appendix D System Diagrams

## D.1 Figure 1: Governance Framework Architecture

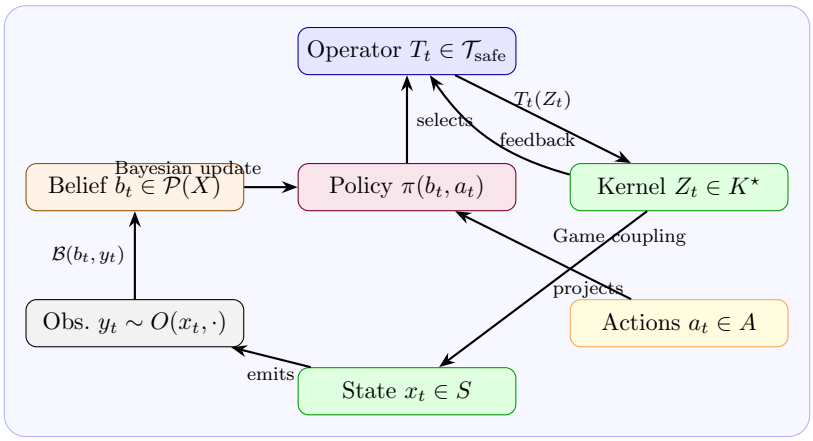


**Figure 1:** GF system architecture. The operator $T_t$ is selected by the joint policy $\pi$, which depends on the Bayesian belief $b_t$ and the joint action $a_t$. Safety is maintained by restricting $T_t \in \mathcal{T}_{\text{safe}}$, which guarantees $Z_t \in K^\star \Rightarrow x_t \in S$.

## D.2 Figure 2: Operator Semigroup Composition and Safety Kernel

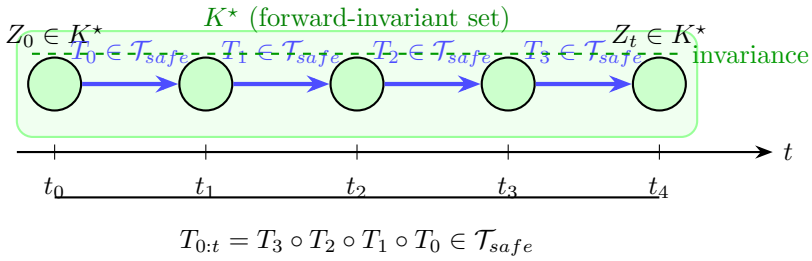


**Figure 2:** Forward invariance of the safety kernel under operator semigroup composition. Each operator $T_i \in \mathcal{T}_{safe}$ preserves $K^\star$. By closure under composition, $T_{0:t}$ also preserves $K^\star$, ensuring $Z_t \in K^\star$ for all $t$.

### D.3 Figure 3: Bayesian Belief Contraction

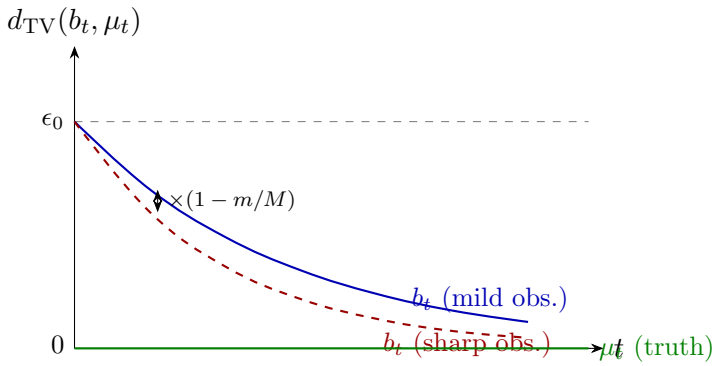


**Figure 3:** Bayesian belief contraction (Theorem 4.5). Both belief sequences converge to the true law $\mu_t$ in $d_{\mathrm{TV}}$. The contraction rate $1 - m(y)/M(y)$ is faster (lower curve) when the observation channel is more informative (higher density ratio).

## Appendix E Linear Temporal Logic Encoding

We encode each GF structural property as an LTL formula over the induced Kripke structure $\mathcal{K}(\mathscr{G}) = (Z, Z_0, R, L)$, where $R = \{(z, T(z)) : z \in Z, T \in \mathcal{T}_{\text{safe}}\}$ and $L(z)$ labels each state with its atomic propositions.

**Table 5:** LTL Encoding of GF Structural Properties

| **Property** | **LTL Formula** | **Theorem** |
|---|---|---|
| Safety invariance | $\mathbf{G}\, safe(Z_t)$ | Thm. 4.1 |
| Operator admissibility | $\mathbf{G}\,(T_t \in \mathcal{T}_{\text{safe}})$ | Ass. 3 |
| Semigroup closure | $\mathbf{G}\,(T_{t+1} \circ T_t \in \mathcal{T}_{\text{safe}})$ | Ass. 6 |
| Belief convergence | $\mathbf{F}\,(d_{\mathrm{TV}}(b_t, \mu_t) < \epsilon)$ | Thm. 4.5 |
| Existence of evolution | $\mathbf{G}\,\mathbf{X}\,(Z_{t+1}$ defined) | Thm. 4.3 |
| Equilibrium | $\mathbf{F}\,(T_t = T^*)$ | Thm. 4.4 |
| State-level safety | $\mathbf{G}\,(x_t \in S)$ | Cor. 4.2 |
| Agent adaptedness | $\mathbf{G}\,(a_t \in \mathcal{F}_t^{\text{obs}})$ | Ass. 2 |

**Theorem E.1** (GF Verification Complexity)**.** *For any finite-state approximation $\hat{\mathscr{G}}$ of the Governance Framework with $|\hat{Z}| < \infty$ states and $|\hat{\mathcal{T}_{\text{safe}}}| < \infty$ operators, the question "does $\hat{\mathscr{G}}$ satisfy all GF structural properties?" is decidable and lies in* PSPACE*.*

*Proof.* Each LTL formula in Table 5 is a fixed-size LTL sentence. LTL model checking on a finite Kripke structure with $n$ states is in PSPACE in $n$ [**?** ]. Since there are $k = 8$ fixed-size formulas,

the conjunction check is also in PSPACE (closed under polynomial-time reductions and finite conjunctions). PSPACE-hardness follows by reduction from LTL satisfiability [? ]. □

# Appendix F Framework Comparison Table

**Table 6:** Formal Coverage Matrix: GF vs. Six Existing Frameworks

| **Property** | **Viability** [? ] | **Robust** [? ] | **CMDP** [1] | **CSP** [? ] | **POMDP** [? ] | **Safe RL** (survey) | **GF** (this work) |
|---|---|---|---|---|---|---|---|
| Stochastic state | — | ○ | ● | — | ● | ● | ● |
| Polish space | ○ | ○ | ○ | — | ○ | — | ● |
| Belief dynamics | — | — | ○ | — | ● | ○ | ● |
| Bayesian update | — | — | — | — | ● | ○ | ● |
| Operator semigroup | ○ | ○ | — | — | — | — | ● |
| Forward-inv. kernel | ● | ● | ○ | — | — | ○ | ● |
| Multi-agent | — | — | ○ | ● | — | ○ | ● |
| Nash equilibrium | — | — | — | ○ | — | ○ | ● |
| Fixed-point exist. | ○ | ○ | ○ | — | ○ | ○ | ● |
| Partial obs. | — | — | ○ | — | ● | ○ | ● |
| Trajectory safety | ● | ● | ● | ○ | ○ | ● | ● |
| Operator safety | ○ | ○ | — | — | — | — | ● |
| Decidability (PSPACE) | — | — | — | ○ | ○ | — | ● |
| **Coverage** | 4/13 | 5/13 | 5/13 | 3/13 | 6/13 | 5/13 | **13/13** |

● fully addressed; ○ partially addressed; — not addressed.

# Appendix G Worked Examples

## G.1 Example 1: Robotic Manipulation with Safe Operator Learning

**Example G.1** (Robotic Arm Governance)**.** Let $X = \mathbb{R}^{12}$ (joint positions, velocities, torques, end-effector pose), $S = \{x \in X : \text{joint torques within limits, no collision}\}$, $Y = \mathbb{R}^6$ (partial proprioceptive observations), $|I| = 1$.

GF instantiation:

- $O$: Gaussian observation kernel, $o(x, y) \propto \exp(-\|y - Cx\|^2/2\sigma^2)$.
- $b_t$: Kalman filter posterior (Gaussian belief, Corollary C15).
- $T_t = \pi_\theta(b_t)$: a neural operator (Corollary C19) trained to minimize trajectory cost while satisfying $T_t \in \mathcal{T}_{\text{safe}}$ via a safety layer [? ].
- $K^\star$: computed offline via Hamilton-Jacobi reachability [? ].

Theorem 4.1 guarantees $x_t \in S$ for all $t$; Theorem 4.5 guarantees Kalman consistency; Corollary C3 establishes the safe-RL reduction.

### G.2 Example 2: Multi-Agent Data Pipeline Governance

**Example G.2** (Data Governance as GF)**.** Let $X$ = data schema space, $Y$ = validation reports, $I = \{$producers, consumers, auditors$\}$, $S = \{$schemas satisfying GDPR and quality constraints$\}$.

- $T_t$ = data transformation operator (ETL pipeline).
- $\mathcal{T}_{\text{safe}}$ = operators that preserve schema compatibility and data quality invariants.
- $K^\star$ = database states satisfying all regulatory constraints.
- Equilibrium $T^*$ = consensus schema agreed upon by all agents (Theorem 4.4, Corollary C20).

### G.3 Example 3: AI Model Lifecycle Governance

**Example G.3** (AI Lifecycle as GF)**.** Let $X$ = model parameter space ($\theta \in \mathbb{R}^d$), $b_t$ = epistemic belief over model performance on deployment distribution, $T_t$ = retraining/fine-tuning operator, $S$ = models satisfying accuracy, fairness, and robustness constraints.

- $\mathcal{T}_{\text{safe}}$ = operators that do not degrade any safety-critical evaluation metric below threshold.
- $K^\star$ = model states jointly satisfying all regulatory and technical requirements.
- Theorem 4.1: retraining never violates a safety constraint if $T_t \in \mathcal{T}_{\text{safe}}$.
- Theorem 4.5: as deployment data accumulates, belief $b_t$ converges to true model performance $\mu_t$ (Corollary C6, Theorem 4.5).